Lab 3: Component Modelling — Blood Bank Management System

Architecture Selection: We chose a Layered Architecture for the Blood Bank Management System.

Reasons for the choice: First, the system has distinct responsibilities: the user portal handles interactions, authentication manages access, donor and donation services maintain donor records, the request service validates and matches blood requests, and the inventory component tracks available stock. Separating these responsibilities improves maintainability and makes each part easier to test. Second, blood requests depend on accurate inventory data. Dedicated request and inventory components can coordinate stock checks and updates while the database maintains a consistent record of requests, donations, and blood units.

Security advantage: A separate Authentication & Roles component can enforce role-based access for donors, hospital users, and staff. Protected operations such as updating inventory or fulfilling a request should be restricted to authorized roles, and database access should use validated queries.

Performance benefit: Separating inventory lookup from request handling allows blood availability to be checked directly without mixing it with user-interface logic. Indexed database queries for blood group, availability, and expiry can reduce lookup time and help staff respond faster to urgent requests.

Interfaces and data flow: The diagram shows seven components and more than four interactions: user portal to authentication, portal to donor management, donor/donation to request processing, request service to inventory and database, inventory updates, and notifications. Lollipop symbols denote provided interfaces and socket symbols denote required interfaces; labelled connectors show the data or service exchanged. This is a proposed component-level architecture based on common blood-bank workflows; confirm the module names against the team's implementation before submission.